

Guillera’s Series for $1/\pi^2$: A Critical Analysis for Decimal Digit Extraction

Joseph Babcanec
Benedict College
`joseph.babcanec@benedict.edu`

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Abstract

We analyze the feasibility of replacing the Chudnovsky series in the Hybrid Chudnovsky–Newton digit extractor (Algorithm 2 of our prior work) with one of Guillera’s series for $1/\pi^2$. The research log conjectured that Guillera’s fastest series gives “ ~ 29 digits per term,” yielding a $\sim 2\times$ speedup. We show this claim is incorrect on both counts. A careful identification of the actual Guillera formulas reveals that the fastest proved series for $1/\pi^2$ converges at approximately 3.01 decimal digits per term—far slower than the Chudnovsky series at 14.18 digits per term. Moreover, even in a hypothetical scenario where a Guillera-type formula matched Chudnovsky’s convergence rate, the extra Newton phase required to recover π from $1/\pi^2$ (inverse square root at $3M(n)$ per step vs. simple inversion at $2M(n)$ per step) would make the Guillera hybrid 33% *slower*, not faster. We provide complete derivations of three candidate inversion routes, prove quadratic convergence for each, compute operation counts, and carry out a detailed complexity comparison. The Chudnovsky hybrid remains optimal among all known hypergeometric approaches.

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iteration

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1 Guillera’s Series: Identification and Exact Coefficients

1.1 Background and motivation

In the research log accompanying our hybrid Chudnovsky–Newton digit extractor, we noted:

“Guillera’s fastest series gives ~ 29 digits per term for $1/\pi^2$. This would require two inversions (square root + inversion). The CRT decomposition carries over since $M_k = (6k)!/((3k)!(k!)^3)$ appears in Guillera’s formula too. Base changes from 640320^3 to 4608 (smaller, cleaner). Could give $\sim 2\times$ additional speedup over our hybrid.”

We now show that **every claim in this passage is incorrect**: the convergence rate, the factorial structure, the base, and the speedup estimate. The source of the error was a conflation of the Chudnovsky series structure (a ${}_3F_2$ hypergeometric series for $1/\pi$ using the factorial ratio $(6k)!/((3k)!(k!)^3)$, connected to class-number-1 imaginary quadratic fields) with the Guillera series structure (a ${}_5F_4$ hypergeometric series for $1/\pi^2$ using fifth powers of Pochhammer symbols, connected to Calabi–Yau differential equations).

1.2 The actual Guillera formulas

Jesús Guillera discovered his series for $1/\pi^2$ in 2002–2003 [1]. They are *not* Ramanujan–Sato type series (which are ${}_3F_2$ series for $1/\pi$), but rather ${}_5F_4$ hypergeometric series. The four principal formulas, using standard Pochhammer symbol notation $(a)_k = a(a+1)\cdots(a+k-1)$, are:

Formula G1 (Guillera 2002, proved by Guillera via WZ):

$$\frac{128}{\pi^2} = \sum_{k=0}^{\infty} \left(-\frac{1}{1024} \right)^k \frac{(\frac{1}{2})_k^5}{(1)_k^5} (820k^2 + 180k + 13). \quad (1)$$

Convergence ratio: $|z| = 1/1024$. Digits per term: $\log_{10}(1024) \approx \mathbf{3.01}$.

Formula G2 (Guillera 2002, proved by Guillera via WZ):

$$\frac{8}{\pi^2} = \sum_{k=0}^{\infty} \left(-\frac{1}{4} \right)^k \frac{(\frac{1}{2})_k^5}{(1)_k^5} (20k^2 + 8k + 1). \quad (2)$$

Convergence ratio: $|z| = 1/4$. Digits per term: $\log_{10}(4) \approx \mathbf{0.60}$.

Formula G3 (Guillera 2002, proved by Guillera via WZ):

$$\frac{32}{\pi^2} = \sum_{k=0}^{\infty} \left(\frac{1}{16} \right)^k \frac{(\frac{1}{4})_k (\frac{1}{2})_k^3 (\frac{3}{4})_k}{(1)_k^5} (120k^2 + 34k + 3). \quad (3)$$

Convergence ratio: $|z| = 1/16$. Digits per term: $\log_{10}(16) \approx \mathbf{1.20}$.

Formula G4 (Guillera 2011, proved by Zudilin):

$$\frac{48}{\pi^2} = \sum_{k=0}^{\infty} \left(\frac{27}{64} \right)^k \frac{(\frac{1}{3})_k (\frac{1}{2})_k^3 (\frac{2}{3})_k}{(1)_k^5} (74k^2 + 27k + 3). \quad (4)$$

Convergence ratio: $|z| = 27/64$. Digits per term: $\log_{10}(64/27) \approx \mathbf{0.37}$.

These are **all** the proved Guillera-type formulas for $1/\pi^2$ as of 2026. Formula G1 is the fastest, at approximately 3.01 digits per term.

1.3 Why the convergence rates are what they are

For a ${}_5F_4$ series $\sum c_k z^k$ with Pochhammer parameters approaching 1, the ratio of consecutive terms satisfies

$$\frac{c_{k+1}}{c_k} \rightarrow z \quad \text{as } k \rightarrow \infty.$$

More precisely, for formula G1 with $(1/2)_k^5/(1)_k^5$:

$$\frac{(1/2)_{k+1}}{(1)_{k+1}} \cdot \frac{(1)_k}{(1/2)_k} = \frac{k+1/2}{k+1} \rightarrow 1,$$

so the fifth power contributes a factor approaching $1^5 = 1$, and the geometric decay is controlled entirely by $z = -1/1024$. This confirms $|a_{k+1}/a_k| \rightarrow 1/1024$, giving $\log_{10}(1024) \approx 3.01$ digits of precision per term.

1.4 Comparison with the Chudnovsky series

The Chudnovsky series for $1/\pi$ is:

$$\frac{1}{\pi} = \frac{12}{C^{3/2}} \sum_{k=0}^{\infty} (-1)^k \frac{(6k)!}{(3k)!(k!)^3} \cdot \frac{L_k}{C^{3k}}, \quad C = 640320, \quad L_k = 13591409 + 545140134k. \quad (5)$$

The factorial ratio $M_k = (6k)!/((3k)!(k!)^3)$ grows as (by Stirling):

$$M_k \sim \frac{\sqrt{2}}{(2\pi)^{3/2}} \cdot \frac{1728^k}{k^{3/2}}. \quad (6)$$

(Derivation in Appendix A.) The convergence rate per term is:

$$\log_{10} \left(\frac{C^3}{1728} \right) = \log_{10} \left(\frac{640320^3}{1728} \right) = 3 \log_{10} \left(\frac{640320}{12} \right) = 3 \log_{10}(53360) \approx 14.18 \text{ digits/term}.$$

Series	Computes	Digits/term	Type
Chudnovsky	$1/\pi$	14.18	${}_3F_2$, Ramanujan–Sato
Guillera G1	$1/\pi^2$	3.01	${}_5F_4$
Guillera G3	$1/\pi^2$	1.20	${}_5F_4$
Guillera G2	$1/\pi^2$	0.60	${}_5F_4$
Guillera G4	$1/\pi^2$	0.37	${}_5F_4$

The Chudnovsky series converges $4.7\times$ faster per term than the fastest Guillera formula.

1.5 Why there is no fast Guillera formula with $M_k = (6k)!/((3k)!(k!)^3)$

The factorial ratio $M_k = (6k)!/((3k)!(k!)^3)$ arises in Ramanujan–Sato type series, which are ${}_3F_2$ hypergeometric series related to modular forms on $\Gamma_0(N)$ for small N . These series compute $1/\pi$, not $1/\pi^2$. The Chudnovsky formula corresponds to the unique class-number-1 discriminant $d = -163$ (the largest Heegner number), which produces the maximal base $C = 640320$ and the fastest convergence rate in this family.

Guillera’s series for $1/\pi^2$ belong to a *different* mathematical family: ${}_5F_4$ hypergeometric series related to Calabi–Yau differential equations. These use Pochhammer products like $(\frac{1}{2})_k^5/(1)_k^5$, not the factorial ratio $(6k)!/((3k)!(k!)^3)$. There is no “ $1/\pi^2$ analogue” of the Chudnovsky series with the same factorial structure and a faster convergence rate. The research log’s claim that “ $M_k = (6k)!/((3k)!(k!)^3)$ appears in Guillera’s formula too” is false.

1.6 Source of the “ ~ 29 digits per term” error

The most likely origin of this erroneous claim is a misreading of the following (correct) observation: the Chudnovsky series provides ~ 14.18 digits per term for $1/\pi$, and $1/\pi^2 = (1/\pi)^2$, so one might naively expect that “squaring” the series would “double” the convergence rate to ~ 28.36 digits per term. This reasoning is fallacious: convergence rate is a property of the *series*, not of the quantity being computed. Squaring a power series does not double its radius of convergence or its per-term digit yield. The Cauchy product of two copies of the Chudnovsky series would indeed give a series for $1/\pi^2$, but its convergence rate would remain ~ 14.18 digits per term (with doubled computational cost per term due to the convolution).

2 Double Inversion: From $1/\pi^2$ to π

Despite the negative conclusion above, the double-inversion problem is mathematically interesting and may arise in other contexts. We analyze three routes from an approximation $S \approx 1/\pi^2$ to π .

2.1 Route A: Newton reciprocal, then Newton square root

Phase A1: From $S \approx 1/\pi^2$, compute $\pi^2 = 1/S$ via Newton's method for $f(y) = 1/y - S$:

$$y_{j+1} = y_j(2 - S \cdot y_j). \quad (7)$$

Phase A2: From π^2 , compute $\pi = \sqrt{\pi^2}$ via Heron's method:

$$z_{j+1} = \frac{1}{2} \left(z_j + \frac{\pi^2}{z_j} \right). \quad (8)$$

Lemma 2.1 (Convergence of Route A). *Both phases converge quadratically.*

Phase A1: Let $e_j = y_j - \pi^2$. Then

$$e_{j+1} = -\frac{e_j^2}{\pi^2}. \quad (9)$$

Phase A2: Let $\epsilon_j = z_j - \pi$. Then

$$\epsilon_{j+1} = \frac{\epsilon_j^2}{2(\pi + \epsilon_j)}. \quad (10)$$

Proof. **Phase A1.** Write $y_j = \pi^2 + e_j$ and $S = 1/\pi^2$:

$$y_{j+1} = (\pi^2 + e_j) \left(2 - \frac{\pi^2 + e_j}{\pi^2} \right) = (\pi^2 + e_j) \left(1 - \frac{e_j}{\pi^2} \right) = \pi^2 - \frac{e_j^2}{\pi^2}.$$

Phase A2. Write $z_j = \pi + \epsilon_j$ and $y = \pi^2$:

$$z_{j+1} = \frac{1}{2} \left(\pi + \epsilon_j + \frac{\pi^2}{\pi + \epsilon_j} \right) = \frac{(\pi + \epsilon_j)^2 + \pi^2}{2(\pi + \epsilon_j)} = \frac{2\pi^2 + 2\pi\epsilon_j + \epsilon_j^2}{2(\pi + \epsilon_j)} = \pi + \frac{\epsilon_j^2}{2(\pi + \epsilon_j)}.$$

Both errors satisfy $|e_{j+1}| = O(e_j^2)$, confirming quadratic convergence. $\square$

Operation count per Newton step:

- Phase A1: $S \cdot y_j$ costs $M(n)$; $y_j \cdot (2 - S y_j)$ costs $M(n)$. Total: $2M(n)$.
- Phase A2 (Heron): requires π^2/z_j , which is a division costing $\sim 3M(n)$ if implemented via Newton reciprocal plus multiplication. Total: $\sim 3M(n)$.

Route A total: $5M(n)$ per combined Newton step.

2.2 Route B: Direct Newton for $\pi = 1/\sqrt{S}$

From $S = 1/\pi^2$, we want $\pi = S^{-1/2}$. Newton's method for $f(w) = 1/w^2 - S$:

$$w_{j+1} = \frac{w_j}{2}(3 - S \cdot w_j^2). \quad (11)$$

Lemma 2.2 (Convergence of Route B). *Let $\epsilon_j = w_j - \pi$. Then*

$$\epsilon_{j+1} = -\frac{\epsilon_j^2(3\pi + \epsilon_j)}{2\pi^2}. \quad (12)$$

For small ϵ_j , $|\epsilon_{j+1}| \approx \frac{3}{2\pi} \epsilon_j^2$.

Proof. Write $w_j = \pi + \epsilon_j$ and $S = 1/\pi^2$:

$$\begin{aligned} w_{j+1} &= \frac{(\pi + \epsilon_j)}{2} \left(3 - \frac{(\pi + \epsilon_j)^2}{\pi^2} \right) \\ &= \frac{(\pi + \epsilon_j)}{2} \left(3 - 1 - \frac{2\epsilon_j}{\pi} - \frac{\epsilon_j^2}{\pi^2} \right) \\ &= (\pi + \epsilon_j) \left(1 - \frac{\epsilon_j}{\pi} - \frac{\epsilon_j^2}{2\pi^2} \right) \\ &= \pi + \epsilon_j - \epsilon_j - \frac{\epsilon_j^2}{2\pi} - \frac{\epsilon_j^2}{\pi} - \frac{\epsilon_j^3}{2\pi^2} \\ &= \pi - \frac{3\epsilon_j^2}{2\pi} - \frac{\epsilon_j^3}{2\pi^2}. \end{aligned}$$

□

Operation count per Newton step (Route B):

- w_j^2 : one multiplication, $M(n)$.
- $S \cdot w_j^2$: one multiplication, $M(n)$.
- $3 - Sw_j^2$: one subtraction, $O(n)$.
- $w_j \cdot (3 - Sw_j^2)$: one multiplication, $M(n)$.
- Division by 2: bit shift, $O(n)$.

Route B total: $3M(n)$ per Newton step.

2.3 Route C: Newton square root, then Newton inversion

Phase C1: From $S = 1/\pi^2$, compute $\sqrt{S} = 1/\pi$ via Heron's method (cost $\sim 3M(n)$).

Phase C2: From $1/\pi$, compute π via Newton inversion $y_{j+1} = y_j(2 - (1/\pi) \cdot y_j)$ (cost $2M(n)$).

Route C total: $5M(n)$ per combined Newton step (same as Route A).

2.4 Comparison of routes

Route	$M(n)$ per step	Leading error constant	Newton phases
A: reciprocal $\rightarrow$ sqrt	$2 + 3 = 5$	$1/\pi^2, 1/(2\pi)$	2 sequential
B: direct $1/\sqrt{S}$	3	$3/(2\pi) \approx 0.477$	1 unified
C: sqrt $\rightarrow$ reciprocal	$3 + 2 = 5$	$1/(2\pi), 1/\pi$	2 sequential

Proposition 2.3 (Optimal inversion route). *Route B is optimal. It requires only $3M(n)$ per Newton step, versus $5M(n)$ for Routes A and C. All three routes achieve quadratic convergence with the same number of steps $S = \lceil \log_2(N/p_0) \rceil + 1$.*

Proof. Route B combines the reciprocal and square-root operations into a single Newton iteration for $w = S^{-1/2}$, saving $2M(n)$ per step. The larger convergence constant ($3/(2\pi) \approx 0.477$ vs. smaller constants in individual phases) does not affect the number of doubling steps needed, since all routes double precision per step. The cost difference is purely in the per-step multiplication count. $\square$

Remark 2.4. *Route B is the inverse-square-root iteration, the same one used in hardware implementations (e.g., the well-known Quake III “fast inverse square root”). Applied here to arbitrary-precision arithmetic, it computes $\pi = 1/\sqrt{1/\pi^2}$ directly.*

3 Precise Convergence Analysis for Route B

Theorem 3.1 (Full convergence of Route B). *Let $S = 1/\pi^2$ and $w_0 > 0$ with $|w_0 - \pi| < \pi$. Define $w_{j+1} = w_j(3 - Sw_j^2)/2$. Then:*

- (a) (w_j) converges to π .
- (b) $\epsilon_{j+1} = -\epsilon_j^2(3\pi + \epsilon_j)/(2\pi^2)$ where $\epsilon_j = w_j - \pi$.
- (c) For $|\epsilon_j| < \pi/3$: $|\epsilon_{j+1}| < (2/\pi)\epsilon_j^2$.
- (d) If $|\epsilon_0| < 10^{-p_0}$, then $|\epsilon_j| < 10^{-p_0 \cdot 2^j}$ for all $j \geq 1$.

Proof. Parts (a) and (b) follow from Lemma 2.2. For (c): when $|\epsilon_j| < \pi/3$, we have $|3\pi + \epsilon_j| < 10\pi/3$, so

$$|\epsilon_{j+1}| < \frac{\epsilon_j^2 \cdot 10\pi/3}{2\pi^2} = \frac{5\epsilon_j^2}{3\pi} < \frac{2\epsilon_j^2}{\pi}.$$

For (d): with $c = 2/\pi < 1$ and $|\epsilon_0| < 10^{-p_0}$:

$$|\epsilon_j| < c^{2^j-1} \cdot 10^{-p_0 \cdot 2^j} < 10^{-p_0 \cdot 2^j},$$

since $c < 1$ implies $c^{2^j-1} < 1$. The number of correct digits doubles at each step. $\square$

4 Variable-Precision Schedule

4.1 Chudnovsky hybrid (baseline)

Starting precision: $p_0^C \approx 14$ digits (one Chudnovsky term). Newton steps: $S_C = \lceil \log_2(N/14) \rceil + 1$. Cost per step: $2M(P_j)$.

4.2 Guillera hybrid (using G1, the fastest)

Starting precision: $p_0^G \approx 3$ digits (one Guillera G1 term). Newton steps: $S_G = \lceil \log_2(N/3) \rceil + 1$.
 Cost per step: $3M(P_j)$.

Note that $p_0^G < p_0^C$, so more Newton steps are needed for Guillera.

4.3 Step count comparison for $N = 1000$

Step j	Chudnovsky		Guillera G1	
	Precision	Terms	Precision	Terms
0	14	1	3	1
1	28	2	6	2
2	56	4	12	4
3	112	8	24	8
4	224	16	48	16
5	448	32	96	32
6	896	64	192	64
7	1005	71	384	128
8			768	256
9			1005	334

Chudnovsky: 7 Newton steps at $2M$ each.

Guillera G1: 9 Newton steps at $3M$ each.

The Guillera hybrid requires *more* Newton steps (because p_0 is smaller) *and* each step is more expensive (3 multiplications vs. 2).

5 Complexity Comparison

5.1 Total Newton phase cost

For a variable-precision Newton iteration starting at p_0 digits, ending at N digits, with c multiplications per step and schoolbook multiplication $M(n) = \Theta(n^2)$:

$$T_{\text{Newton}} = c \sum_{j=1}^S M(p_0 \cdot 2^j) = c \sum_{j=1}^S (p_0 \cdot 2^j)^2 = c p_0^2 \sum_{j=1}^S 4^j = c p_0^2 \cdot \frac{4(4^S - 1)}{3}. \quad (13)$$

Since $p_0 \cdot 2^S \approx N$, we have $p_0^2 \cdot 4^S \approx N^2$, so:

$$T_{\text{Newton}} \approx \frac{4c}{3} N^2. \quad (14)$$

Proposition 5.1 (Newton cost independence of p_0). *The total Newton phase cost depends on the per-step multiplication count c but **not** on the starting precision p_0 (to leading order). This is because the geometric sum $\sum 4^j$ is dominated by its last term.*

This is a crucial observation: even though Guillera G1 starts from fewer digits ($p_0 = 3$ vs. 14), requiring more Newton steps, the extra early steps are cheap and contribute negligibly to the total.

Chudnovsky Newton cost:

$$T_N^C \approx \frac{4 \cdot 2}{3} N^2 = \frac{8}{3} N^2 \approx 2.67 N^2.$$

Guillera Newton cost:

$$T_N^G \approx \frac{4 \cdot 3}{3} N^2 = 4 N^2.$$

5.2 Total series evaluation cost

At each Newton step j , the series must be evaluated to P_j digits of precision using binary splitting. The binary splitting cost for T terms of a hypergeometric series at precision P is $O(M(P) \log T)$ in the standard analysis.

For Chudnovsky at step j : $T_j = \lceil P_j/14.18 \rceil$ terms.

For Guillera G1 at step j : $T_j = \lceil P_j/3.01 \rceil$ terms — approximately $4.7\times$ more terms.

However, since binary splitting reduces the evaluation to a single large-number division at cost $O(M(P_j))$, and the variable-precision sum is dominated by the final step, both methods have series cost $\approx (4/3) N^2$ to leading order with schoolbook multiplication, with Guillera having a modestly larger constant due to the $\log T$ factor.

More precisely, the series cost ratio is:

$$\frac{T_{\text{series}}^G}{T_{\text{series}}^C} \approx \frac{\log(N/3.01)}{\log(N/14.18)} \approx 1 + \frac{\log(14.18/3.01)}{\log(N/14.18)},$$

which approaches 1 as $N \rightarrow \infty$. For $N = 1000$, this ratio is approximately 1.37.

5.3 Total cost comparison

$$T_{\text{Chudnovsky}}(N) \approx \frac{8}{3} N^2 + \frac{4}{3} N^2 = 4 N^2, \tag{15}$$

$$T_{\text{Guillera}}(N) \approx 4 N^2 + \frac{4}{3} N^2 \cdot 1.37 \approx 4 N^2 + 1.83 N^2 = 5.83 N^2. \tag{16}$$

Theorem 5.2 (Guillera is slower). *Using the fastest proved Guillera formula (G1, 3.01 digits/term), the Guillera hybrid digit extractor is approximately*

$$\frac{T_{\text{Guillera}}}{T_{\text{Chudnovsky}}} \approx \frac{5.83}{4} \approx 1.46$$

times slower than the Chudnovsky hybrid. The slowdown comes from two sources:

1. *The Newton inverse-square-root iteration costs $3M(n)$ per step vs. $2M(n)$ for simple inversion (50% overhead in the Newton phase).*
2. *The series converges $4.7\times$ slower, requiring more terms per binary-splitting evaluation (modest overhead in the series phase).*

5.4 Hypothetical analysis: what if a faster Guillera formula existed?

Suppose hypothetically that a Guillera formula for $1/\pi^2$ existed with the same convergence rate as Chudnovsky ($\delta = 14.18$ digits/term). Then the series costs would be identical, and the comparison reduces to Newton costs alone:

$$\frac{T_{\text{Guillera}}^{\text{hyp}}}{T_{\text{Chudnovsky}}} = \frac{4N^2 + (4/3)N^2}{(8/3)N^2 + (4/3)N^2} = \frac{16/3}{4} = \frac{4}{3} \approx 1.33.$$

Even in this best-case hypothetical, the Guillera hybrid is **still 33% slower** due to the inherent cost of the inverse-square-root Newton iteration.

5.5 Can the inverse-square-root iteration be done in $2M(n)$?

No. The iteration $w_{j+1} = w_j(3 - Sw_j^2)/2$ requires:

1. Computing w_j^2 : this is a squaring, which costs $M(n)$ (or $\sim 0.7M(n)$ for algorithms that exploit $a = b$ in multiplication).
2. Computing $S \cdot w_j^2$: one multiplication, $M(n)$.
3. Computing $w_j \cdot (3 - Sw_j^2)$: one multiplication, $M(n)$.

Steps 1–3 are all essential; none can be eliminated without breaking the iteration. Even with asymmetric multiplication (Karatsuba, FFT), squaring is at best $\sim 0.7M(n)$, reducing the total to $\sim 2.7M(n)$ —still more than $2M(n)$.

5.6 What convergence rate would break even?

For a Guillera formula with convergence rate δ_G to match the Chudnovsky hybrid, we need the series-phase savings to exactly compensate the Newton-phase overhead. The break-even condition is:

$$\frac{4}{3}N^2 \cdot \frac{\log(N/\delta_G)}{\log(N/14.18)} + 4N^2 = 4N^2.$$

This gives $\delta_G = \infty$: even with *infinitely fast* series convergence (zero series cost), the Newton phase alone costs $4N^2$, which equals the *entire* Chudnovsky cost of $4N^2$. So a Guillera hybrid can never strictly beat a Chudnovsky hybrid.

6 CRT Decomposition Analysis

6.1 Structure of Guillera coefficients

The Guillera formulas use Pochhammer products, not the factorial ratio $M_k = (6k)!/((3k)!(k!)^3)$. For G1:

$$\frac{(\frac{1}{2})_k^5}{(1)_k^5} = \frac{(\frac{1}{2})_k^5}{(k!)^5} = \left(\frac{(2k)!}{4^k \cdot k!} \right)^5 \cdot \frac{1}{(k!)^5} = \frac{(2k)!^5}{4^{5k} \cdot (k!)^{10}}.$$

This can also be written as $\binom{2k}{k}^5 / 4^{5k}$, where $\binom{2k}{k}$ is the central binomial coefficient. Combined with $z^k = (-1/1024)^k = (-1)^k / 2^{10k}$:

$$\frac{(\frac{1}{2})_k^5}{(1)_k^5} \cdot z^k = (-1)^k \frac{(2k)!^5}{2^{10k} \cdot 2^{10k} \cdot (k!)^{10}} = (-1)^k \frac{\binom{2k}{k}^5}{2^{10k}}.$$

6.2 CRT decomposition for Guillera

The CRT decomposition from our prior work was designed for the Chudnovsky factorial ratio $M_k = (6k)! / ((3k)!(k!)^3)$, where the recurrence $M_{k+1}/M_k = R(k)$ has denominators involving $(3k+1)(3k+2)(3k+3)(k+1)^3$.

For the Guillera coefficient $F_k = \binom{2k}{k}^5$, the recurrence is:

$$\frac{F_{k+1}}{F_k} = \frac{\binom{2k+2}{k+1}^5}{\binom{2k}{k}^5} = \left(\frac{(2k+2)!/(k+1)!^2}{(2k)!/k!^2} \right)^5 = \left(\frac{(2k+1)(2k+2)}{(k+1)^2} \right)^5 = \left(\frac{2(2k+1)}{k+1} \right)^5.$$

The denominator is $(k+1)^5$. The “bad primes” for CRT are those dividing $(j+1)$ for some $j < k$, i.e., all primes $p \leq k$. The linear term (quadratic polynomial) $P(k) = 820k^2 + 180k + 13$ must be analyzed modulo these primes.

Proposition 6.1 (CRT viability for Guillera). *The CRT decomposition is viable for the Guillera coefficients $F_k \bmod P(k)$, with:*

1. *Clean part: all primes $p > k$ dividing $P(k)$, where modular recurrence works.*
2. *Bad part: product of primes $p \leq k$ dividing $P(k)$, computed via direct binomial evaluation.*

The bad part is larger than in the Chudnovsky case (since all primes $\leq k$ are potentially bad, vs. only primes dividing specific denominator factors for Chudnovsky), but remains manageable: $D_k = O(k^{1+\epsilon})$ in bit-length.

6.3 Quadratic vs. linear modular term

In Chudnovsky, $L_k = 13591409 + 545140134k$ is linear, so $L_k \equiv 0 \pmod{p}$ has exactly one solution mod p . For Guillera, $P(k) = 820k^2 + 180k + 13$ is quadratic, so $P(k) \equiv 0 \pmod{p}$ has at most 2 solutions mod p . This at most doubles the conflict frequency, with negligible impact on overall complexity.

7 Alternating Signs and Binary Splitting

7.1 All proved Guillera formulas are alternating

Formulas G1 and G2 have $z < 0$ (alternating signs). Formula G3 has $z = 1/16 > 0$ (non-alternating). Formula G4 has $z = 27/64 > 0$ (non-alternating). However, the non-alternating formulas G3 and G4 have the *slowest* convergence rates (1.20 and 0.37 digits/term respectively).

The fastest formula G1 is alternating ($z = -1/1024$). This is not coincidental: the alternating-sign structure is connected to the analytic continuation of the ${}_5F_4$ function, and the formulas with $z < 0$ converge faster because they evaluate the hypergeometric function at a point further from the radius of convergence on the negative real axis.

7.2 Does alternation cause problems for binary splitting?

No. Binary splitting handles signs in the standard way: the Pochhammer ratio $(k+1/2)/(k+1)$ at each step is positive, and the sign $(-1)^k$ is tracked as $\text{sign}(z)^k$. All intermediate quantities in the binary splitting recursion are exact integers (possibly negative), and the final rational approximation is computed by a single division. This is identical to how the Chudnovsky series handles its $(-1)^k$ factor.

7.3 Is there a non-alternating Guillera formula fast enough for digit extraction?

Among proved formulas: no. The fastest non-alternating formula (G3) achieves only 1.20 digits/term, which is $11.8\times$ slower than Chudnovsky. A non-alternating formula would enable direct modular extraction (BBP-style) without binary splitting, but no such formula with competitive convergence exists.

This parallels the situation for $1/\pi$: the Chudnovsky series is also alternating, and no non-alternating Ramanujan-type formula with competitive convergence is known. The fundamental reason is that the fastest-converging formulas arise from evaluating modular functions at specific CM points, and these evaluations inherently produce alternating series.

8 Net Assessment and Conclusions

8.1 Summary of findings

1. **The “ ~ 29 digits per term” claim is false.** The fastest proved Guillera formula for $1/\pi^2$ achieves 3.01 digits/term, which is $4.7\times$ slower than Chudnovsky’s 14.18 digits/term.
2. **The factorial ratio $(6k)!/((3k)!(k!)^3)$ does not appear in Guillera’s formulas.** Guillera’s series use Pochhammer products like $(\frac{1}{2})_k^5/(1)_k^5$, which are central-binomial-coefficient structures, not the Ramanujan–Sato factorial ratio.
3. **The “base changes to 4608” claim is false.** The base for G1 is $1024 = 2^{10}$, not 4608.
4. **Even with matching convergence rates, Guillera would be slower.** The inverse-square-root Newton iteration costs $3M(n)$ per step vs. $2M(n)$ for simple inversion, a 50% overhead that cannot be compensated by any improvement in series convergence.
5. **The optimal inversion route is Route B** (direct $w_{j+1} = w_j(3 - Sw_j^2)/2$), saving $2M(n)$ per step over the two-phase alternatives (Routes A and C).
6. **All fast Guillera formulas are alternating.** The non-alternating variants (G3, G4) converge too slowly for practical digit extraction.
7. **The CRT decomposition carries over in modified form.** The Pochhammer recurrence for $(\frac{2k}{k})^5$ has denominator $(k+1)^5$, creating a denser set of bad primes than the Chudnovsky case, but the decomposition remains viable.
8. **The Chudnovsky hybrid remains optimal** among all known hypergeometric series approaches for π digit extraction.

8.2 Corrected research log entry

The entry in Section 4.2 of the research log should read:

The fastest *proved* Guillera formula for $1/\pi^2$ (formula G1, 2002) achieves approximately 3.01 digits per term via the ${}_5F_4$ hypergeometric function with Pochhammer structure $(\frac{1}{2})_k^5/(1)_k^5$ and geometric ratio $z = -1/1024$. This is $4.7\times$ slower per term than Chudnovsky’s 14.18 digits per term. Moreover, recovering π from $1/\pi^2$ requires

a Newton inverse-square-root iteration at $3M(n)$ per step, which is 50% more expensive than the simple Newton inversion ($2M(n)$) used to recover π from $1/\pi$ in the Chudnovsky hybrid. A complete complexity analysis shows the Guillera hybrid would be approximately $1.46\times$ *slower* than the Chudnovsky hybrid, not faster. The Chudnovsky hybrid remains optimal.

8.3 Remaining open directions

The analysis above closes the Guillera direction but suggests two productive alternatives:

1. **Higher-discriminant Ramanujan–Sato formulas for $1/\pi$.** The Chudnovsky series uses discriminant $d = -163$, the largest Heegner number. There are exactly 9 Heegner numbers $(-1, -2, -3, -7, -11, -19, -43, -67, -163)$, and $d = -163$ yields the maximum convergence rate. No faster Ramanujan-type series for $1/\pi$ can exist within this family—the Chudnovsky formula is already optimal.
2. **Reducing the Newton inversion cost.** The hybrid’s bottleneck is the Newton phase at $2M(n)$ per step. Any method that avoids inversion entirely (computing π directly rather than $1/\pi$) would eliminate this cost. However, as noted in our research log (Section 4.3), all known direct- π series involve irrational prefactors, which destroy modular arithmetic.

A Stirling Verification for $M_k = (6k)!/((3k)!(k!)^3)$

By Stirling’s formula:

$$\begin{aligned}(6k)! &\sim \sqrt{12\pi k} \cdot (6k/e)^{6k}, \\ (3k)! &\sim \sqrt{6\pi k} \cdot (3k/e)^{3k}, \\ (k!)^3 &\sim (2\pi k)^{3/2} \cdot (k/e)^{3k}.\end{aligned}$$

Therefore:

$$M_k \sim \frac{\sqrt{12\pi k} \cdot 6^{6k} k^{6k} e^{-6k}}{\sqrt{6\pi k} \cdot 3^{3k} k^{3k} e^{-3k} \cdot (2\pi k)^{3/2} \cdot k^{3k} e^{-3k}} = \frac{\sqrt{2}}{(2\pi k)^{3/2}} \cdot \frac{6^{6k}}{3^{3k}}.$$

Since $6^6/3^3 = 46656/27 = 1728$:

$$\boxed{M_k \sim \frac{\sqrt{2}}{(2\pi)^{3/2}} \cdot \frac{1728^k}{k^{3/2}}}.$$

The Chudnovsky series term decays as:

$$|a_k| \sim \frac{1728^k}{C^{3k}} \cdot k \cdot \text{const} = \frac{k}{(C^3/1728)^k} \cdot \text{const}.$$

With $C^3/1728 = 640320^3/1728 = 53360^3$:

$$\text{digits/term} = \log_{10}(53360^3) = 3 \log_{10}(53360) \approx 3 \times 4.727 = 14.18.$$

References

- [1] J. Guillera, About a new kind of Ramanujan-type series, *Experiment. Math.* **12** (2003), 507–510.
- [2] J. Guillera, Generators of some Ramanujan formulas, *Ramanujan J.* **11** (2006), 41–48.
- [3] J. Guillera, Hypergeometric identities for 10 extended Ramanujan-type series, *Ramanujan J.* **15** (2008), 219–234.
- [4] J. Guillera, A new Ramanujan-like series for $1/\pi^2$, *Ramanujan J.* **26** (2011), 369–374.
- [5] G. Almkvist and J. Guillera, Ramanujan-like series for $1/\pi^2$ and string theory, *Experiment. Math.* **21** (2012), 223–234.
- [6] W. Zudilin, Quadratic transformations and Guillera’s formulas for $1/\pi^2$, *Math. Notes* **81** (2007), 297–301.
- [7] J. M. Campbell, Three-parameter generalizations of formulas due to Guillera, arXiv:2502.15249, 2025.